

Ahmed Yassine Hachem

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Work Experience

PURA Solutions

AI Data Labeling

June 2024 – July 2024

- Led an AI training initiative, labeling 20,000+ product images with detailed metadata and designing a scalable pipeline that reduced project time by 67%.
- Achieved 98% labeling accuracy across 10,000+ images and collaborated with the CTO on QA for 20,000 labels, improving model accuracy by 20%.

Brokins FR

AI Engineer

June 2025 – July 2025

- Collaborated with a co-worker to build two production ready AI tools from scratch for Brokins, a French insurance company.
- Designed and implemented a RAG chatbot capable of searching across data from numerous French insurance companies and answering instantly
- Built a smart insurance comparator that takes user preferences and recommends the most suitable insurer using AI-based matching
- Handled personally the server deployment, connecting through SSH and managing setup with PUTTY and FileZilla, making sure both tools ran smoothly on Brokins infrastructure

Automotive Team , OTH-AW

AI Engineer Intern

April 2026 – Present

- Fused LiDAR, IMU, and wheel odometry to generate real-time point clouds and 3D mesh reconstructions of tunnel environments; produced grid-based maps for DRL-based wireless repeater placement in Sionna/Gazebo simulations
- Adapted and deployed AI models on edge devices, including real-world data preprocessing and format transformation for model compatibility
- Evaluated sim-to-real gap and fine-tuned AI models on real-world data to improve deployment robustness

Projects

NexusBi

- Designed and built a full-stack BI platform (React/TypeScript, Flask, REST) that ingests heterogeneous tabular data and performs automated statistical preprocessing (IQR outlier removal, imputation, label encoding, feature scaling) followed by LLM-driven relationship discovery and report generation via the Gemini API.
- Engineered a two-stage, prompt-chained AI pipeline that grounds a generated business-intelligence report in independently computed feature-relationship data, with temperature and token-budget tuning per call stage.
- Implemented multi-tenant session isolation and file-hash-based response caching in a stateless-deployment architecture spanning cross-origin frontend and backend hosts.

Queue Intelligence System

- Engineered a real-time multi-object tracking pipeline (YOLOv8 + BoT-SORT) with GPU-accelerated CUDA/FP16 inference and custom polygon-based ROI filtering, reducing identity fragmentation by ~90% in occluded, multi-person retail footage
- Designed a hybrid heuristic-to-LLM correction system where lightweight statistical triggers flag ambiguous tracking events for a Gemini vision agent, which analyzes extracted video segments and returns schema-constrained JSON verdicts to resolve identity ambiguity
- Built a full-stack Streamlit application with an interactive in-browser zone-annotation tool, live video analytics dashboard, and a decoupled precomputed-replay deployment mode, shipped as a public demo on Streamlit Community Cloud

FertiFy

- Built a two-stage multimodal retrieval pipeline: a vision-LLM call converts an image into a structured symptom description, embedded and matched via similarity search, then re-combined with the original image for final generation.
- Abstracted the retrieval layer behind a common interface, decoupling the search/storage implementation from the embedding and generation logic.
- Designed structured prompt schemas constraining Gemini outputs to fixed response formats with explicit confidence-based branching.

Education

Pristini School of AI

Sousse, Tunisia

Bachelor in Applied Artificial Intelligence

Sep 2023 – Jan 2026

Relevant courses: Deep learning | Machine Learning | Advanced Python programming | C | Java | Data Engineering | Data Administration | Computer Vision | Data Mining | Big Data

Ostbayerische Technische Hochschule Amberg-Weiden

Amberg, Germany

Bachelor in Artificial intelligence | Exchange program

March 2026 - Present

Skills

Programming Languages: Python, C, SQL, Java

Robotics & Simulation: ROS 2, LiDAR-Inertial SLAM (FAST-LIO2), PCL, KISS-Matcher

ML/AI: TensorFlow, Keras, scikit-learn, RAG, Multi-Agent Systems, Pandas, NumPy, MLOps, MLflow

Tools: VS Code, Eclipse IDE, AWS, Git, Power BI

Languages: English (C1), French (B2)